PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: Lampoly Etch of Si on Si/SiO_2/Metal/Glass Substrates

Requestor: John Abendroth, David Barton, Prof. Jennifer Dionne **Date:** 04/19/19

Non-SNF staff, please complete the following:

Badger ID: dbarton, abend. Email: dbarton@stanford.edu, abend@stanford.edu

Phone: 612-222-8442 (David), 850-459-9319 (John)

Company name/PI: Jennifer Dionne

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups and the Stanford Chemical Safety Data Base sections on this website to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

   Composition of substrates for Si etch by Lampoly: Glass substrate/Ta (3 nm)/Pt (3 nm)/[Pt (0.7 nm)/Co (0.6 nm)]x2/Pt (3.7 nm)/Ta (3 nm)/SiO_2 (30 nm)/Si (130 nm).

   Glass substrates are: double side polished soda lime glass from University Wafer (ID number : 1631). The wafer thickness is around 615 um, although 550 um is mentioned in University City website.

   These substrates are prepared outside of SNF (see 2). Growth recipe: Metal layers will be grown by a DC magnetron sputtering system with base pressure of < $3x10^{-8}$ Torr. The SiO_2 layer will be grown by RF sputtering (100 W) and the Si layer will be grown by DC sputtering (100 W). Depositions will be performed at room temperature in an Ar gas atmosphere (2.7 mTorr).

2. **Vendor/manufacturer info:** address and phone number, website URL.

   Substrates are grown at the Center for Memory and Recording Research, University of California, San Diego, La Jolla, CA 92093.

   https://cmrr.ucsd.edu

   Director: Prof. Eric Fullerton, (858) 534-9639

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

   We have designed a new nanophotonic device which requires silicon features made with highly anisotropic etching and smooth sidewalls. The structure involves a glass substrate, a thin (15 nm) metal stack, deposited SiO_2 (30 nm) and a 130nm Si top layer. We require both anisotropic etching of the silicon layer as well as chemistries to stop on the oxide layer. After discussing with Usha, we decided that lampoly was the best tool for our process needs. However, the Cl_2 and HBr chemistries in the Si etch will attack most semi-clean metals. This could contaminate the chamber as well as destroy the device. Thus, the reason for the process request is for permission to use the tool with the substrates described above because they have ~15 nm metal layer. Importantly, however, the metal layer is separated from the Si to be etched and surface by 30 nm SiO_2, preventing direct exposure of the metal to the plasma in the chamber when substrates are properly mounted and etched (no sample edges exposed, see below).

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that f the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

   Please note that the process flow included here has been discussed with Usha.

   The sample is cleaned in Acetone followed by Isopropyl Alcohol, surface functionalized (SURPASS 4000), and coated in ma-N 2403 resist. The JEOL is used to define patterns on the silicon. This step is done in SNC.

   The piece will be mounted to a dedicated carrier silicon wafer by a combination of Santovac oil and Kapton tape and not either or. Diffusion pump oil is used to provide better thermal conduction between sample and substrate and Kapton tape is needed for preventing the sides (both glass and metals) to be exposed during processing.

   The piece on the carrier wafer is etched in lampoly and timed to stop on the oxide layer.

Because the sample will stop on the oxide layer, the etch should not reach the metal layer. The sample will be mounted in the center of the wafer to avoid any issues with the clamp in lampoly.

Following the etch, the samples will be removed from the carrier wafer and the resist removed with Acetone.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

   One 8 x 8.5 mm substrate will be loaded in the tool per use.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

   Substrates will not be stored in SNF.

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

   N/A

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: